

University Regulation RAG Assistant

Project Report

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1. Overview

This notebook implements a Retrieval-Augmented Generation (RAG) assistant designed to answer questions about a university's academic regulations, policies, syllabi, and administrative guidelines. Rather than relying on a language model's built-in knowledge, the system retrieves relevant passages from a locally indexed set of official PDF documents and grounds every generated answer in that retrieved context, citing the specific source file and page number for each claim.

The corpus consists of institutional documents belonging to Vignan University (VFSTR), covering regulations across programs (B.Tech, M.Tech, MBA, MCA, Law, Pharmacy, and others), academic policies (scholarships, admissions, research, grievance redressal), codes of conduct, and course/syllabus structures.

2. Objective

Build a question-answering system that:

- Ingests a large, heterogeneous set of university PDFs and preserves document/page-level metadata.
- Retrieves the most relevant passages for a natural-language question using semantic search.
- Generates a factual answer strictly grounded in retrieved context, explicitly declining to answer when the context is insufficient.
- Always attaches a source list (filename + page number) so answers can be verified against the original regulation text.

3. Architecture & Pipeline

The notebook follows a standard four-stage RAG pipeline, implemented entirely in Google Colab:

3.1 Data ingestion

PDFs are read from a Google Drive folder using LangChain's PyPDFLoader, which preserves each page as a separate document with its source filename and page number as metadata. Files that are corrupted or password-protected are skipped rather than failing the whole run.

3.2 Chunking

Loaded pages are split into overlapping chunks using RecursiveCharacterTextSplitter (chunk_size = 1000 characters, chunk_overlap = 200 characters), balancing retrieval precision against the context needed for coherent answers.

3.3 Embedding & indexing

Each chunk is embedded with the sentence-transformers/all-MiniLM-L6-v2 model (via HuggingFaceEmbeddings) and stored in a local FAISS vector index, enabling fast similarity search entirely within the Colab runtime — no external vector database is required.

3.4 Retrieval & generation

At query time, the top-k most similar chunks (default k = 4–8) are retrieved via similarity_search_with_score. Their text and source metadata are assembled into a single context block and inserted into a strict grounding prompt sent to google/gemini-2.5-flash through the OpenRouter API. The prompt instructs the model to answer only from the supplied context and to state explicitly when an answer cannot be found, rather than guessing.

4. Dataset Statistics

The indexing step processed the following volume of material:

Metric	Value
Source PDF documents in Drive folder	201
Total pages loaded across all files	21,880
Text chunks created (chunk_size=1000, overlap=200)	56,648
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector index	FAISS
Generation model	google/gemini-2.5-flash
Max output tokens per answer	1,000

Document category breakdown (by chunk count), computed by classifying filenames into four buckets — Policies, Regulations, Syllabus/Curriculum, and Other Guidelines:

Category	Chunk Count	Approx. Share
Regulations	45,471	80.3%
Other Guidelines	6,399	11.3%

Syllabus/Curriculum	4,588	8.1%
Policies	190	0.3%

5. Evaluation — Test Queries

Three representative regulatory questions were run against the assistant to validate end-to-end behavior:

#	Test Question	Sources Retrieved
1	Eligibility criteria for securing a scholarship	BCA R26 Regulations; SCHOLARSHIPS POLICY; M.A English R26 Regulations; R26 Regulations B.Tech (4 sources)
2	How the university promotes industry linkages and application-oriented research	V3_R25 Book; Service Rules (VFSTR); Code of Conduct (faculty) (3 sources)
3	Disciplinary measures for examination malpractice	Disciplinary_Action_-_Malpractices.pdf, pages 1, 4 and 5 (3 sources, single document)

5.1 Sample findings

- Scholarship eligibility: correctly synthesized rules from four different regulation/policy documents, noting that merit scholarships are unavailable to students in spill-over semesters.
- Industry linkages: pulled a clean, itemized list of initiatives (consultancy, collaborative research, faculty industry training, incubators) from three source documents.
- Examination malpractice: produced a detailed, clause-by-clause breakdown of punishments (expulsion, cancellation of performance, debarment, police referral) with page-level citations, and correctly surfaced the appeals process to the Vice-Chancellor.

A separate ad-hoc query on B.Tech CSE (AI&ML) Category-B tuition fees was also verified, correctly returning ₹3,80,000/- per year tuition and a ₹20,000/- non-refundable first-year admission fee, sourced from the official fee-structure and admission-policy PDFs — demonstrating the system also handles administrative/financial lookups, not just regulatory text.

6. Key Design Choices & Safeguards

- Hallucination protection: the prompt template forces the model to answer only from retrieved context and to say “I cannot find the answer in the provided documents” rather than invent one.
- Source attribution: every answer is paired with a machine-readable list of (filename, page) tuples pulled directly from document metadata, not generated by the model.

- Resilient ingestion: per-file try/except handling means a single corrupted or encrypted PDF cannot halt processing of the other 200 documents.

7. Limitations & Suggested Improvements

- Ephemeral index: the FAISS store is rebuilt in-memory every run; persisting it (FAISS `save_local/load_local`) would avoid re-embedding all 56,648 chunks on every session.
- No re-ranking: retrieval relies purely on embedding similarity; adding a re-ranking step (e.g., cross-encoder or MMR) could improve precision for ambiguous queries.
- Filename-based categorization: the Policies/Regulations/Syllabus split in the visualization is a keyword heuristic on filenames and will misclassify files that don't contain those substrings.
- Single embedding model: all-MiniLM-L6-v2 is compact and fast but lower-capacity than larger retrieval-tuned embeddings; a stronger embedding model could improve recall on nuanced regulatory language.
- No automated evaluation metric: correctness of answers is currently checked manually against three sample questions rather than a held-out QA benchmark.

8. Conclusion

The notebook delivers a working, source-grounded RAG assistant capable of answering diverse questions — regulatory, procedural, and financial — across 201 university documents and nearly 22,000 pages, with every answer traceable to a specific file and page. The design prioritizes factual grounding and transparency over open-ended generation, making it suitable as a first-line reference tool for students, faculty, or administrative staff navigating dense institutional regulations.